

14.9 Appendices

Appendix 14.A: Equivalence of Peak Distortion and Zero-Forcing Criterion

The peak distortion criterion and zero-forcing criterion are equivalent, as shown in [Proakis and Salehi 2005]. Let the concatenation of discrete-time channel and equalizer be an equivalent filter with impulse response

$$q_i = \sum_{n=-\infty}^{\infty} e_n f_{i-n} . \quad (14.43)$$

At the i -th sampling instant, the sample value is

$$\hat{c}_i = q_0 c_i + \sum_{n \neq i} c_n q_{i-n} + \sum_{n=-\infty}^{\infty} e_n n_{i-n} . \quad (14.44)$$

The first term represents the desired symbol, while the second one is the intersymbol interference, and the third the (correlated) noise. The maximum ISI is

$$D = \sum_{n \neq 0} |q_n| = \sum_{n \neq 0} \left| \sum_{k=-\infty}^{\infty} e_k f_{n-k} \right| . \quad (14.45)$$

An equalizer with an infinite number of weights can always eliminate the ISI (as long as the transfer function is finite at all considered frequencies). Elimination of ISI means

$$q_i = \sum_{n=-\infty}^{\infty} e_n f_{i-n} = \begin{cases} 1 & i = 0 \\ 0 & i \neq 0 \end{cases} . \quad (14.46)$$

A z -transformation of this equation reads

$$Q(z) = E(z)F(z) = 1. \quad (14.47)$$

The equalizer that minimizes the maximum ISI is thus the inverse of the channel in the transform domain, i.e., a zero-forcing equalizer.

Appendix 14.B: Derivation of the Mean-Square-Error Criterion

We are searching for a filter that minimizes:

$$MSE = E \{ |\varepsilon_i|^2 \} = E \{ \varepsilon_i \varepsilon_i^* \} \quad (14.48)$$

The estimated sequence $\{\hat{c}_i\}$ can be written as:

$$\hat{c}_i = \sum_{n=-K}^K e_n u_{i-n} = \mathbf{e}^T \mathbf{u}_i = \mathbf{u}_i^T \mathbf{e} \quad (14.49)$$

where $\mathbf{u}_i = (u_i, u_{i-1}, \dots)^T$.

The MSE can thus be written as:

$$\begin{aligned} MSE &= E \{ (c_i - \mathbf{u}_i^T \mathbf{e}) (c_i^* - \mathbf{e}^\dagger \mathbf{u}_i^*) \} \\ &= E \{ c_i c_i^* \} - E \{ c_i \mathbf{e}^\dagger \mathbf{u}_i^* \} - E \{ c_i^* \mathbf{u}_i^T \mathbf{e} \} + \mathbf{e}^\dagger E \{ \mathbf{u}_i^* \mathbf{u}_i^T \} \mathbf{e} \end{aligned} \quad (14.50)$$

This equation contains four expected values:

- The expected value $E \{ c_i c_i^* \}$ is the variance of c_i (assuming that c_i is zero-mean) $\sigma_S^2 = E \{ c_i c_i^* \}$.
- The expected value $\mathbf{p} = E \{ \mathbf{u}_i^* c_i \}$ is the cross-correlation between the received vector and the transmit sequence.
- The expected value $E \{ c_i^* \mathbf{u}_i^T \}$ is:

$$\mathbf{p}^\dagger = E \{ c_i^* \mathbf{u}_i^T \} \quad (14.51)$$

- The expected value $E \{ \mathbf{u}_i^* \mathbf{u}_i^T \}$ is the correlation matrix of the received signal:

$$\mathbf{R} = E \{ \mathbf{u}_i^* \mathbf{u}_i^T \} \quad (14.52)$$

so that (for uncorrelated transmit sequences) $R_{ki} = \zeta_{|k-i|}$.

With these definitions, the MSE can be written as:

$$MSE = \sigma_S^2 - \mathbf{e}^\dagger \mathbf{p} - \mathbf{p}^\dagger \mathbf{e} + \mathbf{e}^\dagger \mathbf{R} \mathbf{e} \quad (14.53)$$

In order to find the optimum weight vector $\mathbf{e}_{\text{opt}}$ we set to 0 the derivative of the MSE with respect to $\mathbf{e}$. This gives us the discrete form of the “Wiener-Hopf” equation:

$$\mathbf{R} \mathbf{e}_{\text{opt}} = \mathbf{p} \quad (14.54)$$

The optimum weight vector is thus $\mathbf{e}_{\text{opt}} = \mathbf{R}^{-1} \mathbf{p}$. For its determination, we need the correlation matrix of the received signal, as well as the cross-correlation between the received signal and the transmit signal.

Appendix 14.C: The Recursive-Least-Squares Algorithm

In the following, we give the details of the RLS algorithm. We define the instantaneous error for the n -th iteration step as

$$\varepsilon_i = c_i - \mathbf{e}_n^T \mathbf{u}_i \quad (14.55)$$

and minimize

$$\Upsilon_n = \sum_{i=0}^n \beta(n, i) |\varepsilon_i|^2. \quad (14.56)$$

Here, $\beta(n, i)$ is a *forgetting factor*, which is usually chosen as

$$\beta(n, i) = (\mu_{\text{RLS}})^{n-i}, \quad i = 1, 2, \dots, n \quad (14.57)$$

where μ_{RLS} is a constant $0 < \mu_{\text{RLS}} < 1$ (for $\mu_{\text{RLS}} = 1$, the algorithm reduces to the usual LS algorithm). We thus want to minimize

$$\Upsilon_n = \sum_{i=1}^n (\mu_{\text{RLS}})^{n-i} |\varepsilon_i|^2. \quad (14.58)$$

We first define the weighted correlation matrix of the receive vector

$$\Psi_n = \sum_{i=0}^n (\mu_{\text{RLS}})^{n-i} \mathbf{u}_i^* \mathbf{u}_i^T \quad (14.59)$$

and the crosscorrelation between desired output and the receive signal

$$\theta_n = \sum_{i=0}^n (\mu_{\text{RLS}})^{n-i} \mathbf{u}_i^* c_i. \quad (14.60)$$

The optimum receive vector is then obtained from the Wiener criterion

$$\hat{\mathbf{e}}_n = \Psi_n^{-1} \theta_n. \quad (14.61)$$

On the other hand, those correlations can also be written in a recursive form:

$$\begin{aligned} \Psi_n &= \mu_{\text{RLS}} \Psi_{n-1} + \mathbf{u}_n^* \mathbf{u}_n^T, \\ \theta_n &= \mu_{\text{RLS}} \theta_{n-1} + \mathbf{u}_n^* c_n. \end{aligned} \quad (14.62)$$

Inversion of Eq. (14.62) gives

$$\Psi_n^{-1} = \mu_{\text{RLS}}^{-1} \Psi_{n-1}^{-1} - \mu_{\text{RLS}}^{-1} \frac{\Psi_{n-1}^{-1} \mathbf{u}_n^* \mathbf{u}_n^T \Psi_{n-1}^{-1}}{\mu_{\text{RLS}} + \mathbf{u}_n^T \Psi_{n-1}^{-1} \mathbf{u}_n^*}. \quad (14.63)$$

Using the definitions

$$\mathbf{P}_n = \Psi_n^{-1} \quad (14.64)$$

and

$$\mathbf{k}_n = \frac{\mathbf{P}_{n-1} \mathbf{u}_n^*}{\mu_{\text{RLS}} + \mathbf{u}_n^T \mathbf{P}_{n-1} \mathbf{u}_n^*} \quad (14.65)$$

we can reformulate Eq. (14.63) according to

$$\mathbf{P}_n = \mu_{\text{RLS}}^{-1} \mathbf{P}_{n-1} - \mu_{\text{RLS}}^{-1} \mathbf{k}_n \mathbf{u}_n^T \mathbf{P}_{n-1} \quad (14.66)$$

and thus obtain the following recursive equation for the weight vectors

$$\hat{\mathbf{e}}_n = \hat{\mathbf{e}}_{n-1} + \mathbf{k}_n [c_n - \mathbf{u}_n^T \hat{\mathbf{e}}_{n-1}]. \quad (14.67)$$

In order to avoid singularities during the initialization, Eq. (14.59) has to be slightly modified, according to

$$\mathbf{\Psi}_n = \sum_{i=0}^n (\mu_{\text{RLS}})^{n-i} \mathbf{u}_i^* \mathbf{u}_i^T + \delta_{\text{INI}} \mu_{\text{RLS}}^n \mathbf{I} \quad (14.68)$$

where δ_{INI} is a small positive number, and $\mathbf{I}$ is the identity matrix.

The RLS algorithm thus consists of the following steps::

1. Set $\mathbf{P}_0 = (1/\delta_{\text{INI}})\mathbf{I}$ and $\hat{\mathbf{e}}_0 = \mathbf{0}$.
2. Increase the index n of the received vector by one. With the new received vector $\mathbf{u}_n$ we compute the new $\mathbf{k}_n$ from Eq. (14.65) and the new $\mathbf{P}_n$ from Eq. (14.66).
3. Using Eq. (14.67) we compute the new weight vector $\hat{\mathbf{e}}_n$.
4. Loop back to step 2.